

Valerie Ha

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EDUCATION

The University of Chicago

Chicago, IL

Bachelor of Science in Molecular Engineering (Quantum track) and Computational Neuroscience | GPA: 3.5/4.0

Expected, June 2028

- Relevant Coursework: Linear Algebra, Mathematics for Quantum Mechanics, Computational Modeling of Brain Circuits, Modeling and Signal Analysis, Principles of Engineering Analysis, Molecular Engineering Thermodynamics, Physics I–III, General Chemistry I–III

Phillips Academy Andover

Andover, MA

High School Diploma | GPA: 5.7/6.0

June, 2024

EXPERIENCE

Marlene Cohen Research Lab - UChicago Department of Neurobiology

Chicago, IL

Undergraduate Researcher

February, 2025 – Present

- Analyze electrophysiology and behavioral datasets from macaque experiments to investigate neural dynamics underlying cognitive control.
- Develop recurrent neural network (GRU-based) models in PyTorch to study adaptive decision-making under probabilistic reward-switching environments. Train and analyze reinforcement learning agents and behavior-cloned neural networks on these dynamic tasks.
- Build analysis pipelines for PCA, shared latent-space projections, and decoders of hidden-state dynamics across network architectures.
- Participating in the Training Program in Computational Neuroscience for Next Generation Neuroscientists (TPCN).

Sihong Wang Research Lab - Pritzker School of Molecular Engineering

Chicago, IL

Undergraduate Researcher

January, 2025 – Present

- Engineer flexible polymer-semiconductor devices for biocompatible signal transmission by formulating precursor solutions and spin-coating multilayer films (hole-injection, emissive, and electron-transport layers) onto Ag-nanowire electrodes.
- Characterize thin-film uniformity and performance via fluorescence spectroscopy using a tin-oxide-coated integrating sphere to maximize light emission, iteratively optimizing material compositions and processing parameters to improve device yield and reliability.
- Iteratively optimize material compositions and fabrication parameters to improve device yield, reliability, and signal transmission.

CoThera - Digital healthcare platform

Chicago, IL

Software Engineer Intern

June, 2025 – September, 2025

- Conducted customer discovery at hospitals - interviewing 50 patients and caregivers to synthesize pain points, and translate 5 key findings.
- Developed all mobile app wireframes and front-end code using React Native, ensuring consistency across screens and user flows.
- Collaborated closely with backend engineers to integrate APIs and UI for key features such as live transcript display and glossary support.
- Prototyped and benchmarked multiple speech-to-text models (WhisperX, Apple) for accuracy and latency, then deployed the chosen model into the mobile app, handling real-time audio streaming, chunked transcript updates, and live medical visit summarization.

Stanford Rural Education Action Program

Stanford, CA

Research Paper Co-Author

June, 2022 – August, 2023

- Developed early childhood assessment tools using Bayley-III test data to improve educational access for 10,000+ children in rural China
- Applied advanced psychometric modeling (Item Response Theory, Exploratory Factor Analysis, Principal Component Analysis) to optimize assessments by removing 130 low-variance test items (variance < 0.01, factor loading < 0.4)

LEADERSHIP AND PROJECTS

University of Chicago Varsity Track and Field

Chicago, IL

Long Jump and Triple Jump

October, 2024 – Present

- Dedicate 10+ hours weekly to practices and lifts during the fall, 15+ hours weekly to practices and meets during the winter and spring

UChicago Neurotech Club

Chicago, IL

Co-Founder and Project Lead

October, 2025 – Present

- Design and lead a hands-on curriculum teaching members to acquire, preprocess, and analyze EEG data (LSL streaming, signal filtering)
- Lead EEG headset hardware initiative to train members in end-to-end BCI development

Mutual Fund Probabilistic Forecasting Platform - React, FastAPI, XGBoost, Python

Chicago, IL

Goldman Sachs Emerging Leaders Series Engineering Program

January, 2026 – April, 2026

- Collaborated on a multi-university engineering team to develop a full-stack financial forecasting platform integrating probabilistic machine learning models with Monte Carlo simulation for long-term investment analysis
- Implemented stochastic simulations using Gaussian sampling and bootstrap resampling of market returns for uncertainty estimation
- Developed frontend visualization tools using React, Recharts, Vite for interactive exploration of probabilistic investment trajectories

Phillips Academy Varsity Basketball

Andover, MA

Co-Captain and All-NEPSAC

November, 2020 – March, 2024

- Pioneered the team's first structured lifting program by designing workout regimen with gym supervisor and led off-season team practices
- Organized 5k runs to raise \$3000+ for children with special needs and Morgan's Message, increasing the team's community engagement

SKILLS AND ADDITIONAL INFORMATION

Languages: English, Spanish, Mandarin, Cantonese

Computer: Python, JavaScript, MATLAB, HTML, Stata, Python, CAD (Fusion 360), Adobe Illustrator

Technical: Spin coating, Fluorescence Spectroscopy, Soldering, Laser cutting, Woodworking, Arduino, RaspberryPi

Interests: Quantum computing, photonics, neurotechnology, hiking, architecture, videography, basketball